HELIXOR MEDTECH

DATA QUALITY AUDIT & DELIVERY PERFORMANCE — TECHNICAL REPORT

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Dataset period	April - September 2026
Dataset size	1,000 raw rows
Report date	11 November 2026
Colab Notebook Link	https://colab.research.google.com/drive/1MDC0FdMTQ02K-_hl6eN-u_fJPI2TlXV6?usp=sharing

Business Context & Problem Statement

Helixor MedTech is a medical device distribution company operating across Europe. The company distributes medical products to hospitals, clinics, laboratories, and medical distributors through multiple warehouses.

The company's distribution operations involve order processing, warehouse dispatch, transportation, and delivery to customers.

Helixor MedTech is facing operational challenges related to late deliveries and incomplete shipments.

These issues may affect customer satisfaction and overall delivery performance.

Therefore, this project aims to analyze shipment data, identify data quality issues, and evaluate delivery performance to determine where operational problems occur.

The analysis will support better decision-making and help the Operations Team identify areas for improvement.

Dataset Profiling & Data Type Review

ISSUE ID	COLUMN	CURRENT DATA TYPE	EXPECTED DATA TYPE
DA-01	promised_delivery_date	Object	Date
DA-02	dispatch_date	Object	Date
DA-03	actual_delivery_date	Object	Date
DA-04	order_date	Object	Date
DA-05	quantity_delivered_units	Object	Whole Number

Data Quality Audit Summary

Issue ID	Field / Area	Issue Type	Count	Example ID
DQ - 01	Warehouse	Missing Value	18	SHP-26-00114
DQ - 02	Promised_delivery_date	Missing Value	6	SHP-26-00320
DQ - 03	actual_delivery_date	Missing Value	261	SHP-26-00223
DQ - 04	quantity_delivered_units	Missing Value	131	SHP-26-00670
DQ - 05	Notes	Blank Value	371	

Duplicate Data

Issue ID	Shipment ID	Occurrences	Row Index	Match Type
DUP - 1	SHP-26-00019	2	58 - 85	Exact duplicate
DUP - 2	SHP-26-00080	2	410 - 443	Exact duplicate
DUP - 3	SHP-26-00072	2	54 - 289	Exact duplicate
DUP - 4	SHP-26-00128	2	682 - 810	Exact duplicate
DUP - 5	SHP-26-00184	2	643 - 954	Exact duplicate
DUP - 6	SHP-26-00245	2	647- 883	Exact duplicate
DUP - 7	SHP-26-00301	2	664 - 991	Exact duplicate
DUP - 8	SHP-26-00357	2	339 - 382	Exact duplicate
DUP - 9	SHP-26-00414	2	144 - 167	Exact duplicate
DUP - 10	SHP-26-00476	2	147 - 277	Exact duplicate

Invalid & Inconsistent Values.

Issue ID	Column	Issue Type	Count	Example Shipment ID	Status
DQ - 01	quantity_delivered_units	Non-numeric Value	4	SHP-26-00412	Needs Confirmation
DQ - 02	promised_delivery_date	Invalid Date	2	SHP-26-00166	Needs Confirmation
DQ - 03	actual_delivery_date	Invalid Date	4	SHP-26-00824	Needs Confirmation
DQ - 04	warehouse	Invalid Warehouse	7	SHP-26-00074	Needs Confirmation

Business Rule Violations

Issue ID	Column	Issue Type	Count	Example Shipment ID	Status
BR - 01	quantity_ordered_units /quantity_delivered_units	Overdelivery	4	SHP-26-00157	Confirmed
BR - 02	quantity_delivered_units / damaged_units	Over Damaged	1	SHP-26-00600	Confirmed
BR - 03	dispatch_date / actual_delivery_date	Before Dispatch	4	SHP-26-00073	Confirmed
BR - 04	order_date / dispatch_date	After Dispatch	1	SHP-26-00176	Needs Confirmation
BR - 05	quantity_ordered_units	Negative Value	3	SHP-26-00109	Confirmed

*BR-04 requires confirmation of the original date format before the business rule violation can be verified.

Business Impact Assessment

Issue	Evidence	Affected Analysis	Potential Business Impact	Decision Needed
Overdelivery	4 Shipment	Quantity Fill Rate	May cause financial losses, inventory discrepancies, and inflated fill rate result	Verify shipment records to determine whether the issue is caused by data entry errors or actual overdelivery.
Invalid Delivery Date	2 Invalid promised 4 Invalid actual dates	On-time Delivery	Invalid delivery dates may lead to incomplete or unreliable OTD results, affecting delivery performance evaluation.	Verify incorrect dates against source records and confirm missing or undetermined dates with the Operations Team.
Negative Ordered Quantity	3 Shipments	Ordered quantity analysis	Negative quantities may distort total ordered quantities and lead to inaccurate Fill Rate calculations.	Verify the correct quantities against the original order records before making corrections
Damaged Unit	1 Affected shipment	Damage Rate	May distort damage rates and lead to financial losses if incorrect records are used for returns or replacements.	Cross-check delivered and damaged quantities against delivery and inspection records before making corrections.
Potential Order After Dispatch	1 Affected shipment	Order Processing	An ambiguous order date may lead to inaccurate order-to-delivery lead time calculations and misleading operational performance results.	Verify the original order date and date format against the source system before confirming a business rule violation or correcting the data.
Delivered Without a valid actual delivery date	12 missing actual date 3 Invalid actual date	On time delivery (OTD)	Missing or invalid delivery dates may prevent accurate OTD and delivery lead time calculations, resulting in unreliable delivery performance reporting.	Verify delivery status and actual delivery dates against Proof of Delivery (POD) and source records before correcting the data.

Issue	Evidence	Affected Analysis	Potential Business Impact	Decision Needed
Delivered Shipment with Zero Delivered Quantity	1 Affected shipment	Quantity fill rate	May lead to inaccurate delivery performance reporting, customer complaints, and potential billing discrepancies.	Verify shipment status and delivered quantity against delivery records and Proof of Delivery (POD) before correcting the data.
Exact duplicate shipments	10 duplicate pairs	Shipment Count	Duplicate records may inflate shipment counts and delivered quantities, leading to inaccurate operational performance reporting.	Verify duplicate records against the source system and confirm the deduplication rule before removing any rows.
Warehouse Inconsistencies	7 Affected shipment	warehouse performance analysis	Inconsistent warehouse labels may split shipment records across multiple categories, leading to inaccurate warehouse performance reporting.	Standardize clear capitalization and whitespace inconsistencies. Confirm ambiguous warehouse labels with the Operations Team before making corrections.
Delivery before dispatch	4 Affected shipments	Delivery lead time & on time delivery	Incorrect date sequences may cause negative delivery lead times and unreliable delivery performance reporting.	Verify dispatch and actual delivery dates against shipment records and Proof of Delivery (POD) before correcting the data.
Missing & non-numeric delivered quantity	131 missing value 4 non numeric value	Quantity Fill Rate	Missing or non-numeric quantities may lead to incomplete or inaccurate Fill Rate calculations and misleading delivery performance results.	Review shipment statuses and verify unresolved quantities against source records before applying corrections.
Inconsistent Order Date Format	6 affected records; 4 unambiguous, 2 ambiguous	Order lead Time	Inconsistent date formats may cause incorrect date interpretation, inaccurate lead time calculations, and unreliable monthly reporting.	Standardize 4 unambiguous dates and verify 2 ambiguous dates against source records.

Business Impact Conclusion

The identified data quality issues may affect the accuracy of key performance indicators (KPIs), particularly On-Time Delivery (OTD) and Quantity Fill Rate. These issues may also lead to inventory discrepancies, unreliable audit results, and misleading operational performance evaluations.

Invalid or ambiguous dates may cause inaccurate delivery lead time calculations. Therefore, unresolved data issues should be verified with the Operations Team and relevant source records before corrections are made and final performance reports are published.

Proposed Cleaning Rules

Issue Type	Proposed Action	Auto-fix Safe ?	Confirmation Needed ?	Reason
Missing Data				
promised_delivery_date	Flag missing dates and verify them against original order records.	NO	YES	The correct promised delivery dates cannot be determined from the available data.
actual_delivery_date	Flag missing dates and verify actual delivery dates against Proof of Delivery (POD) and shipment records.	NO	YES	A missing actual date can be legitimate for In Transit or Cancelled records; Delivered records require verification.
quantity_delivered_units	Review shipment statuses and verify missing delivered quantities against delivery records when required.	NO	YES	Missing quantities may be valid for incomplete or cancelled shipments but require verification for completed deliveries.
warehouse	Flag missing warehouse values and verify the original warehouse locations with the Warehouse Operations Team.	NO	YES	The actual warehouse locations cannot be determined from the available data. Incorrect assignments may affect warehouse performance analysis.
Duplicate				
duplicate shipments	Verify duplicate records against the source system and remove excess duplicate rows after approval.	NO (Need Approval)	YES	Although the records are identical, the deduplication rule must be confirmed before removing any rows to prevent unintended data loss.
Invalid & Non-numeric values				
quantity_delivered_units	Flag non-numeric values, review shipment statuses, and verify unresolved quantities against delivery records.	NO	Conditional	The actual delivered quantities are unknown and cannot be safely replaced with numeric values.
promised_delivery_date	Flag invalid dates and verify the correct or agreed delivery dates against original order and delivery scheduling records.	NO	YES	Invalid or undetermined delivery dates cannot be corrected without verified source information.
actual_delivery_date	Review shipment statuses and verify invalid delivery dates against Proof of Delivery (POD) and source records before correcting them.	NO	YES	The correct delivery dates cannot be determined from invalid or unrecorded values without verified source information.

Issue Type	Proposed Action	Auto-fix Safe ?	Confirmation Needed ?	Reason
Warehouse Formatting Inconsistencies (5)	Standardize capitalization and remove extra whitespace.	YES	NO	The warehouse names are identifiable and only differ in formatting.
Ambiguous / Unknown Warehouse Labels (2)	Verify warehouse locations and mappings with the Operations Team..	NO	YES	The correct warehouse locations cannot be confirmed from the available data
Negative Ordered Quantity				
quantity_ordered_units	Verify negative quantities against original sales orders before making corrections.	NO	YES	It is unclear whether the error is caused by an incorrect negative sign or an incorrect quantity value.
Overdelivery				
quantity_ordered_units - quantity_delivered_units	Verify ordered and delivered quantities against sales orders, warehouse records, and Proof of Delivery (POD).	NO	YES	It is unclear whether the discrepancy is caused by incorrect data entry or actual overdelivery.
Damaged Units > Delivered Units				
damaged	Verify damaged and delivered quantities against Quality Control inspection reports and delivery records.	NO	YES	Damaged units exceed delivered units, violating the business rule. The correct quantities cannot be determined without verification.
Delivery Before Dispatch				
dispatch_date actual_delivery_date	Verify dispatch and actual delivery dates against logistics records and Proof of Delivery (POD).	NO	YES	The recorded delivery date occurs before dispatch, violating the expected shipment sequence. The correct dates require verification.
Potential Order After Dispatch				
order_date dispatch_date	Verify the original order date and date format with the Order Management Team before making corrections.	NO	YES	The ambiguous date format prevents confirmation of whether the order was placed before or after dispatch.
Delivered Without A Valid Date				
shipment_status actual_delivery_date	Verify delivery status and actual delivery dates with the Logistics Team using Proof of Delivery (POD) and shipment records.	NO	YES	Shipments are marked as Delivered, but their actual delivery dates are missing or invalid, preventing reliable delivery performance calculations.
Delivered With Zero Quantity				

Issue Type	Proposed Action	Auto-fix Safe ?	Confirmation Needed ?	Reason
shipment_status quantity_delivered_units	Review shipment status, notes, actual delivery date, and Proof of Delivery (POD) to verify the delivered quantity.	NO	YES	The shipment is marked as Delivered, but the recorded delivered quantity is zero. The correct quantity cannot be determined without verification.

Data Cleaning Actions & Validation

Safe data cleaning was performed using a separate working DataFrame (df_clean) while preserving the original dataset (df_raw).

Five warehouse formatting inconsistencies were corrected. Date and quantity fields were converted into analysis-ready data types without overwriting their original values.

During order date conversion, six records with a different date format were identified. Four unambiguous dates were successfully converted, while two ambiguous dates remain unresolved pending source verification.

Final validation confirmed that the row count remained unchanged and all original columns except warehouse were preserved. Unresolved missing values, invalid dates, quantity issues, and potential business rule violations were retained for further investigation rather than automatically corrected.

PART II | TECHNICAL ANALYSIS & RESULTS

HELIXOR MEDTECH • April–September 2026 • Provisional findings

Scope and reporting note

This section extends the original data-quality audit. It records validated notebook outputs and separates confirmed observations from unresolved operational questions. Figures below are provisional because duplicate handling and certain exceptions are not yet approved. The source dataset is a simulated business assignment; the analysis and report are the author's own work.

10 | SAFE CLEANING AND VALIDATION

Method. Keep df_raw unchanged; apply safe conversions and standardizations to df_clean, preserving original date and quantity fields. Both dataframes retained 1,000 records.

Warehouse. Five clear formatting differences were standardized: rotterdam and ROTTERDAM → Rotterdam; lyon → Lyon; trailing spaces removed from Hamburg and Milan. Frankfurt DC, Unknown Hub, and 18 missing warehouse values remain unresolved rather than guessed.

Dates. Separate parsed fields produced: promised dates—8 NaT (6 missing, 2 invalid); actual dates—265 NaT (261 missing, 4 invalid); dispatch dates—0 NaT. Six mixed-format order dates were found: four unambiguous dates parsed and two ambiguous dates not assigned to a month.

Quantities. The numeric delivered-quantity field has 135 unavailable values (131 blank, 4 non-numeric). Unknown values were not replaced with zero.

Shipment status. Standardized clear variations in a new shipment_status_std column: Delivered 511; Cancelled 128; Partial 124; In Transit 121; Exception 115; Awaiting confirmation 1. The last two labels need business confirmation.

Data integrity. No rows were removed; all original fields except the safe warehouse formatting fixes remained unchanged in before/after comparison.

11 | DERIVED FIELDS

Quantity shortfall = ordered units − delivered units. Positive results indicate a numeric shortfall; negative results indicate possible overdelivery. Missing source quantity remains NaN.

Delivery delay = actual delivery date − promised delivery date (days). Values ≤ 0 are on time or early; values > 0 are late. Missing/invalid dates remain NaN.

Completeness across all 1,000 rows: quantity shortfall available for 865, delivery delay for 728, and both for 718. These are calculation-availability counts, NOT clean-shipment counts or KPI denominators.

12 | DELIVERY KPI METHODOLOGY

On-Time Delivery (OTD). Select normalized Delivered status with valid promised and actual dates. Temporarily exclude all flagged exact-duplicate rows and delivery-before-dispatch anomalies; both copies of each duplicate pair are excluded pending the approved deduplication rule.

OTD trace: 511 Delivered → 490 with parseable delivery dates → 10 duplicate-flagged rows and 4 date-order anomalies (no overlap) → 476 eligible for the provisional calculation. On time: 110; late: 366; OTD = 23.11%.

Quantity Fill Rate. Among normalized Delivered rows, require positive ordered quantity and numeric delivered quantity. From 501 candidate rows, flag 10 duplicate rows, 3 overdeliveries, and 1 zero-delivered record; no overlapping flags. The provisional population is 487.

Unit-weighted calculation: 108,864 units recorded delivered ÷ 111,679 units ordered = 97.48% Fill Rate; difference = 2,815 units (2.52%). The difference is not evidence of confirmed lost inventory.

Limitations. OTD and Fill Rate use DIFFERENT subsets. The zero-delivery exclusion and exclusion of both duplicate copies may change final metrics. Do not present either figure as approved operating performance.

13 | PERFORMANCE BY WAREHOUSE AND ROUTE

****Quantity Fill Rate by Warehouse and Route****

The provisional Quantity Fill Rate analysis included 487 eligible shipment records, representing 111,679 ordered units and 108,864 recorded delivered units. The overall unit-weighted Fill Rate was 97.48%, with an arithmetic shortfall of 2,815 units.

Lyon recorded the lowest warehouse-level Fill Rate at 95.35%, while FRA-SE-03 recorded the lowest route-level Fill Rate at 95.78%. These findings identify areas requiring further quantity reconciliation, but do not establish confirmed inventory loss.

****Damaged Units and Damage Rate****

Within the same 487-record eligible population, Milan recorded the highest absolute number of damaged units among warehouses, at 197 units. However, Lyon recorded the highest warehouse-level Damage Rate at approximately 1.00%.

At route level, HAM-DE-03 recorded the highest damaged-unit count (132 units) and the highest Damage Rate (approximately 1.26%).

Damage Rate was calculated as recorded damaged units divided by recorded delivered units, multiplied by 100. These findings should be interpreted cautiously because damaged-unit records and shipment eligibility remain subject to data-quality verification.

****Interpretation****

The lowest-performing warehouse or route differs across On-Time Delivery, Quantity Fill Rate, and Damage Rate. This indicates that delivery timeliness, quantity fulfillment, and recorded product damage should be evaluated separately rather than treated as a single measure of distribution performance.

14 | MONTHLY PIVOT: WAREHOUSE × ORDER MONTH

The monthly pivot uses the month of order_date, NOT delivery or dispatch month. Two of the 476 provisional OTD records have ambiguous order dates, leaving 474 observations for the monthly comparison.

Key observation. Madrid's April order cohort had 0/7 on time, and August had 0/9 on time. Hamburg's OTD moved from 11.76% (2/17) in July to 47.37% (9/19) in August, then to 14.29% (2/14) in September. These cohorts are small, so percentages should not be read as proof of process improvement.

OPTIONAL INVESTIGATION | MADRID

Of Madrid's 55 evaluated shipments, 50 were late. Five late shipments departed after the promised date; the other 45 departed before or on it. For these 45, average available time = 4.78 days, average actual travel time = 9.00 days, and average lateness = 4.22 days. The medians are 5, 10, and 4 days.

Across all six named warehouses, median available days were shorter than median actual travel days. This is consistent with low timeliness, but it does not establish the underlying cause. No confirmed route-level service standard or operational transit benchmark is provided.

EVIDENCE, HYPOTHESES & OPEN QUESTIONS

FACT. The source contains 1,000 raw rows. Under documented temporary filters, provisional OTD = 110/476 (23.11%) and provisional unit-weighted Fill Rate = 108,864/111,679 (97.48%). The order-month pivot contains 474 observations.

INFERENCE. Low timeliness appears across all six named warehouses, while quantity fulfillment appears relatively high in a different eligible population.

HYPOTHESIS. Promised delivery dates may be overly optimistic; transport or dispatch scheduling and Sales–Logistics coordination may contribute. These are not proven causes.

UNRESOLVED. Duplicate-policy approval; date/quantity exceptions; invalid before-dispatch dates; unexpected status labels; ambiguous and missing warehouse data; overdelivery and zero-delivered cases; evidence from POD, order system, QC logs, and route SLA.

FOLLOW-UP QUESTION FOR OPERATIONS. How are customer promised dates set, and are they checked against approved transport times for the relevant routes?

PROVENANCE & REPRODUCIBILITY

Original analytical work: Thoriq Satrio Ambodo — Junior Data & Operations Analyst portfolio case study. The HELIXOR scenario and shipment data are a supplied fictional assignment, not independently proven proprietary data.

Source spreadsheet:
https://docs.google.com/spreadsheets/d/1Oq6B_5q7AKOBximlXzANqHkOADyKqNoP3NnoPxpEUGg/edit

Working notebook:
https://colab.research.google.com/drive/1MDC0FdMTQ02K-_hl6eN-u_fJPI2TlXV6?usp=sharing

Reproducibility checklist: preserve df_raw; rerun the notebook top-to-bottom; verify exclusion rules, KPI numerators and denominators, 476 warehouse/route rows, and the 474-record monthly pivot. The management update will be maintained as a separate, shorter document.

REFERENCE TABLES | VALIDATED PROVISIONAL OUTPUTS

Table C — Monthly OTD by warehouse (rate with sample count in parentheses), Apr–Sep 2026.

Warehouse	Apr	May	Jun	Jul	Aug	Sep
Frankfurt	28.57% (14)	25.00% (8)	20.00% (10)	31.58% (19)	21.05% (19)	15.38% (13)
Hamburg	33.33% (15)	33.33% (6)	26.67% (15)	11.76% (17)	47.37% (19)	14.29% (14)
Lyon	6.25% (16)	14.29% (7)	27.27% (11)	30.00% (10)	21.43% (14)	8.33% (12)
Madrid	0.00% (7)	16.67% (12)	7.69% (13)	12.50% (8)	0.00% (9)	20.00% (5)
Milan	35.00% (20)	18.18% (11)	16.67% (12)	25.00% (12)	27.78% (18)	30.00% (20)
Rotterdam	14.29% (14)	46.67% (15)	12.50% (8)	15.38% (13)	25.00% (8)	37.50% (16)
Unknown	100.00% (2)	0.00% (2)	66.67% (3)	0.00% (2)	0.00% (2)	0.00% (3)

TABLE A | ON-TIME DELIVERY BY WAREHOUSE

Warehouse	Shipments	On time	Late	OTD
Frankfurt	83	20	63	24.10%
Hamburg	87	24	63	27.59%
Lyon	70	12	58	17.14%
Madrid	55	5	50	9.09%
Milan	93	25	68	26.88%
Rotterdam	74	20	54	27.03%
Unknown	14	4	10	28.57%

TABLE B | ON-TIME DELIVERY BY ROUTE

Route ID	Shipments	On time	Late	OTD
HAM-DK-02	49	6	43	12.24%
RTM-UK-05	49	8	41	16.33%
HAM-DE-03	53	9	44	16.98%
MIL-IT-04	47	9	38	19.15%
FRA-SE-03	61	12	49	19.67%
LYO-FR-02	40	10	30	25.00%
MAD-ES-01	34	9	25	26.47%
RTM-DE-01	60	18	42	30.00%
FRA-AT-01	47	15	32	31.91%
RTM-BE-02	36	14	22	38.89%

TABLE D | QUANTITY FILL RATE BY WAREHOUSE

Warehouse	Shipments	Ordered Units	Delivered Units	Fill Rate
Lyon	75	15.998	15.254	95,35%
Hamburg	87	20.571	19.844	96,47%
Rotterdam	78	17.471	17.050	97,59%
Unknown	14	3.259	3.201	98,22%
Madrid	57	13.217	12.984	98,24%
Milan	93	21.787	21.450	98,45%
Frankfurt	83	19.376	19.081	98,48%
TOTAL	487	111.679	108.864	97,48%

Interpretation. Lyon recorded the lowest provisional Quantity Fill Rate at 95.35%, indicating the largest proportional quantity fulfillment gap among the named warehouses. Results are based on 487 eligible shipment records and remain subject to data-quality verification.

TABLE E | QUANTITY FILL RATE BY ROUTE

Route ID	Shipments	Ordered Units	Delivered Units	Fill Rate
FRA-SE-03	62	14.309	13.705	95,78%
FRA-AT-01	48	11.770	11.298	95,99%
RTM-BE-02	38	8.304	8.028	96,68%
HAM-DK-02	51	11.949	11.629	97,32%
HAM-DE-03	52	10.719	10.488	97,84%
LYO-FR-02	43	9.287	9.120	98,20%
MAD-ES-01	37	8.452	8.304	98,25%
MIL-IT-04	49	12.308	12.101	98,32%
RTM-UK-05	49	11.387	11.202	98,38%
RTM-DE-01	58	13.194	12.989	98,45%
TOTAL	**487**	**111.679**	**108.864**	**97,48%**

Interpretation. FRA-SE-03 had the lowest provisional route-level Fill Rate (95.78%; 13,705 of 14,309 units), with a recorded arithmetic shortfall of 604 units. These figures do not prove physical inventory loss.

TABLE F | DAMAGED UNITS & DAMAGE RATE BY WAREHOUSE

Warehouse	Shipments	Delivered Units	Damaged Units	Damage Rate
Milan	93	21.450	197	0,92%
Frankfurt	83	19.081	176	0,92%
Hamburg	87	19.844	163	0,82%
Rotterdam	78	17.050	163	0,96%
Lyon	75	15.254	153	1,00%
Madrid	57	12.984	119	0,92%
Unknown	14	3.201	30	0,94%
TOTAL	**487**	**108.864**	**1.001**	**0,92%**

Interpretation. Milan recorded the largest absolute damaged-unit count (197 units), while Lyon had the highest rate among named warehouses (1.00%). The Unknown category is not a verified warehouse. Damaged units were measured only within the same 487-record provisional Fill Rate subset.

TABLE G | DAMAGED UNITS & DAMAGE RATE BY ROUTE

Route ID	Shipments	Delivered Units	Damaged Units	Damage Rate
HAM-DE-03	52	10.488	132	1,26%
RTM-DE-01	58	12.989	118	0,91%
HAM-DK-02	51	11.629	117	1,01%
FRA-AT-01	48	11.298	108	0,96%
FRA-SE-03	62	13.705	107	0,78%
MIL-IT-04	49	12.101	97	0,80%
LYO-FR-02	43	9.120	88	0,96%
MAD-ES-01	37	8.304	87	1,05%
RTM-UK-05	49	11.202	86	0,77%
RTM-BE-02	38	8.028	61	0,76%
TOTAL	**487**	**108.864**	**1.001**	**0,92%**

Interpretation. HAM-DE-03 recorded the highest absolute damaged-unit count (132 units) and Damage Rate (1.26%) among analyzed routes. Damage Rate = recorded damaged units / recorded delivered units. These are provisional metrics on 487 quantity-eligible shipments; the route identifier alone does not establish a warehouse of origin or the cause of damage.